

# SPBANKLENS 360

Enterprise Banking Operations & Risk Intelligence  
Platform

## Functional Requirements Specification (FRS)

Conceptualized & Developed by  
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Version 1.0 Prototype  
Date: 27 June 2026

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## Version History

| Version | Date        | Author        | Description                                   |
|---------|-------------|---------------|-----------------------------------------------|
| 1.0     | 27 Jun 2026 | Sagarika Paul | Initial FRS for SPBANKLENS 360 v1.0 Prototype |

# 1. Introduction

## 1.1 Purpose

This Functional Requirements Specification (FRS) describes the implemented functionality of SPBANKLENS 360, an Enterprise Banking Operations & Risk Intelligence Platform. It documents every screen, control, workflow and module as currently delivered in Version 1.0 Prototype, and forms the canonical reference for product demonstration, technical handover, training, QA, and audit.

## 1.2 Scope

The FRS covers the Branch, Regional, Central, and Administration portals; the Account 360 module; Knowledge Notebook 2.0; Rule-Based Risk Engine; Early Warning System; Workflow & Task Management; Reminder Engine; Recovery, NPA, Legal, CGTMSE and Inspection modules; Reports; Search; Notifications; and Security. It excludes external Core Banking System (CBS), Loan Origination System (LOS), and General Ledger functionality — SPBANKLENS 360 is a read-only intelligence layer over those systems.

## 1.3 Intended Users

- Branch Officers and Branch Managers — daily operational use.
- Regional Office Supervisors — branch monitoring and escalations.
- Central Office Executives — enterprise governance and analytics.
- System Administrators — RBAC, configuration, and audit.
- Auditors, BAs, QA Engineers, and Implementation Partners.

## 1.4 Definitions

**Risk Health Index (RHI):** Composite score (0-100) summarising the operational risk of a branch or account.

**EWS:** Early Warning System — rule-based alert generator for emerging risk.

**ATR:** Action Taken Report — closure note for an inspection observation.

**CGTMSE:** Credit Guarantee Fund Trust for Micro and Small Enterprises.

**NPA:** Non-Performing Asset, classified per RBI norms (SMA-0/1/2, Sub-Standard, Doubtful, Loss).

**Knowledge Notebook:** Persistent operational memory of an account, surviving officer transitions.

## 1.5 Acronyms

| Acronym | Expansion                             |
|---------|---------------------------------------|
| FRS     | Functional Requirements Specification |
| RBAC    | Role-Based Access Control             |
| CBS     | Core Banking System                   |

| Acronym | Expansion                        |
|---------|----------------------------------|
| LOS     | Loan Origination System          |
| EWS     | Early Warning System             |
| RHI     | Risk Health Index                |
| NPA     | Non-Performing Asset             |
| SLA     | Service Level Agreement          |
| ATR     | Action Taken Report              |
| RO / CO | Regional Office / Central Office |

## 2. System Overview

SPBANKLENS 360 is a read-only Risk Intelligence Layer that aggregates data from CBS, LOS, recovery, legal and inspection systems and presents it through role-specific operational portals. The platform is built on TanStack Start (React 19, Vite 7) with a Cloudflare Workers edge runtime and httpOnly cookie-based session security.

### 2.1 Implemented Portals & Modules

- Branch Portal — Branch Dashboard, Account 360, Knowledge Notebook, Recovery, Workflow Centre.
- Regional Office Portal — Branch Heat Map, Comparison Tool, Escalation Tracker.
- Central Office Portal — Executive Command Centre, National Heat Map, Risk Intelligence Hub, Officer Continuity.
- Administration — User & Role management, Risk thresholds, Escalation SLAs, Audit logs.
- Cross-cutting — Rule-Based Risk Engine (13 rules), EWS, Workflow Centre (9 queues), Reminder Engine, Email Automation simulator, Knowledge Notebook 2.0 (10-section structured notes with version history).

### 2.2 Architectural Principles

- Read-only — no banking transactions are initiated from SPBANKLENS 360.
- Role-based — every screen is gated by RBAC server-side.
- Explainable — every risk score exposes the rules that produced it.
- Scalable — designed for 10,000+ branches and millions of accounts.

### 3. Functional Modules (Index)

Sections 4 through 23 document each implemented functional module using a standard template covering purpose, navigation, roles, inputs, processing logic, validation, outputs, reports, notifications, dependencies, error handling, security, audit, and integration.

| #  | Module                     | Primary Role                |
|----|----------------------------|-----------------------------|
| 4  | Login & Authentication     | All                         |
| 5  | Branch Dashboard           | Branch                      |
| 6  | Regional Office Dashboard  | Regional                    |
| 7  | Central Office Dashboard   | Central                     |
| 8  | Account 360                | Branch / Regional           |
| 9  | Knowledge Notebook         | Branch / Regional / Central |
| 10 | Rule-Based Risk Engine     | Central (Admin)             |
| 11 | Early Warning System       | Branch / Regional / Central |
| 12 | Workflow & Task Management | All                         |
| 13 | Reminder Engine            | All                         |
| 14 | Recovery Module            | Branch / Regional           |
| 15 | NPA Management             | Branch / Regional / Central |
| 16 | Legal Module               | Branch / Central            |
| 17 | CGTMSE Module              | Branch / Central            |
| 18 | Inspection Module          | Regional / Central          |
| 19 | Reports                    | All                         |
| 20 | Administration             | Admin                       |
| 21 | Search                     | All                         |
| 22 | Notifications              | All                         |
| 23 | Security                   | Admin / System              |

## 4. Login & Authentication

### 4.1. Login & Authentication

**Module Purpose:** Authenticate users and establish a server-verified session bound to a role.

**Navigation Path:** / (landing → role selection) → /login

**Accessible Roles:** Branch User, Regional User, Central User, Security/Admin User

**Business Objective:** Provide secure entry to role-specific portals; prevent privilege escalation via client tampering.

**Inputs:** Selected role, username, password (demo credentials displayed on the login panel).

**Processing Logic:** Credentials are validated by a TanStack server function which issues an encrypted, httpOnly, SameSite=Lax session cookie containing role and user identity. The client never sees the raw session token.

**Validation Rules:** Username and password required; role must match issued credentials; locked accounts rejected.

**Outputs:** Authenticated redirect to the role-default landing route (/branch, /regional, /central, /admin).

**Reports Generated:** Authentication audit log (admin only).

**Notifications:** Inline error toast on failure; success silent.

**Dependencies:** Server session middleware, role guard router middleware.

**Error Handling:** Invalid credentials → inline error; session expiry → forced redirect to /login.

**Security Controls:** httpOnly + Secure cookies; server-side role check on every protected route; no role information stored in localStorage.

**Audit Trail:** Login success, failure, and logout events captured with timestamp and IP.

**Integration with Other Modules:** Used by every protected route; supplies role context to AppShell sidebar and route guards.

### 4.2 Logout

Logout invalidates the server session cookie and redirects to the landing page.

Browser back-navigation to protected routes triggers re-authentication.



## 5. Branch Dashboard

### 5.1. Branch Dashboard

**Module Purpose:** Operational cockpit for Branch Officers and Managers showing portfolio, risk, and daily tasks.

**Navigation Path:** /branch

**Accessible Roles:** Branch User, Branch Manager

**Business Objective:** Surface the most critical operational signals (overdue accounts, EWS alerts, recovery actions) on a single screen.

**Inputs:** Branch code (from session), date range filter, portfolio segment filter.

**Processing Logic:** Aggregates loan portfolio mock data (or uploaded data) and renders 16 KPI cards, a Risk Health Index gauge, Portfolio Trend chart and Risk Distribution chart via Recharts.

**Validation Rules:** Filters constrained to permitted branch; date range max 12 months.

**Outputs:** KPI grid, trend lines, alerts panel, quick-action shortcuts to Account 360 and Workflow.

**Reports Generated:** Branch Daily Summary (PDF/XLSX export).

**Notifications:** Top-bar notification bell shows pending tasks and EWS alerts.

**Dependencies:** Branch mock-data engine, RHI calculator, Workflow store.

**Error Handling:** Empty-state cards when data unavailable; chart re-render on resize.

**Security Controls:** Branch user cannot view another branch.

**Audit Trail:** Dashboard view and export events logged.

**Integration with Other Modules:** Drill-down to Account 360, Knowledge Notebook quick-note, Workflow queue.

### 5.2 KPI Catalogue

| KPI                       | Source                   | Refresh        |
|---------------------------|--------------------------|----------------|
| Total Advances            | Portfolio aggregate      | On load        |
| Active Borrowers          | Portfolio aggregate      | On load        |
| Overdue Accounts          | Aging buckets            | On load / auto |
| SMA-0 / 1 / 2             | Days-past-due classifier | On load        |
| NPA %                     | NPA classification       | On load        |
| EWS Alerts (open)         | EWS engine               | Auto           |
| Recovery Target vs Actual | Recovery module          | On load        |
| CGTMSE Coverage %         | CGTMSE register          | On load        |
| Pending Inspections       | Inspection register      | On load        |
| Risk Health Index         | Rule engine              | On load        |

## 6. Regional Office Dashboard

### 6.1. Regional Office Dashboard

**Module Purpose:** Supervisory view of every branch in the region with comparison and escalation tools.

**Navigation Path:** /regional

**Accessible Roles:** Regional User, Regional Head

**Business Objective:** Identify outlier branches, monitor escalation SLAs, and drive corrective action.

**Inputs:** Region code (session), branch filter, risk band filter.

**Processing Logic:** Computes branch-level RHI, ranks branches, plots a Heat Map (colour-graded by risk), and exposes pairwise comparison.

**Validation Rules:** Region scope enforced server-side.

**Outputs:** Heat map, branch ranking table, escalation queue, regional KPI strip.

**Reports Generated:** Regional Portfolio Excel export, Escalation Aging report.

**Notifications:** SLA-breach badges on overdue escalations.

**Dependencies:** xlsx 0.20.3 export, branch mock-data engine.

**Error Handling:** Graceful empty-state when a branch reports no data.

**Security Controls:** Regional user blocked from other regions.

**Audit Trail:** Drill-down and export events logged.

**Integration with Other Modules:** Drill into Branch Dashboard and Account 360; pushes escalations into Workflow Centre.

## 7. Central Office Dashboard

### 7.1. Executive Command Centre

**Module Purpose:** Enterprise-wide visibility for Central Office executives.

**Navigation Path:** /central

**Accessible Roles:** Central User, Executive, Admin

**Business Objective:** Provide a single-pane view of national health, regional concentration, governance indices, and inspection ATR status.

**Inputs:** Region/zone filter, sector filter, period filter.

**Processing Logic:** National Health Index aggregates regional RHIs weighted by exposure; geographic and sector concentration computed from portfolio data.

**Validation Rules:** Period restricted to last 36 months.

**Outputs:** National Heat Map, sector concentration treemap, governance index gauges, ATR tracker.

**Reports Generated:** Executive MIS Pack (XLSX), Regulatory Calendar.

**Notifications:** Threshold-breach alerts on national KPIs.

**Dependencies:** Risk Intelligence store, Phase-8 data engine.

**Error Handling:** Charts degrade to tables on render failure.

**Security Controls:** Central-only; admin sub-tabs gated by Admin role.

**Audit Trail:** All exports and threshold changes logged.

**Integration with Other Modules:** Drill-down to Regional and Branch dashboards; feeds Risk Intelligence Hub.

## 8. Account 360

### 8.1. Account 360

**Module Purpose:** Single CRM-style view of a loan account combining customer, exposure, risk, recovery, legal, inspection, and knowledge data.

**Navigation Path:** /branch/account/:accountId

**Accessible Roles:** Branch, Regional, Central (read scoped to role)

**Business Objective:** Eliminate context-switching across systems during account review.

**Inputs:** Account ID (URL param) or global search selection.

**Processing Logic:** Fetches mock or uploaded data, renders 10 contextual sections in a tabbed layout, and computes inline risk dials.

**Validation Rules:** Account ID must exist and be within the user's permitted scope.

**Outputs:** 10 panels (see 8.2), early-warning strip, risk dial, downloadable account dossier.

**Reports Generated:** Account Dossier (PDF).

**Notifications:** EWS triggers visible at the top of the page.

**Dependencies:** Knowledge Notebook store, Risk Engine, Recovery store.

**Error Handling:** 404 panel for unknown account; section-level skeleton loaders.

**Security Controls:** Read-only; no edit controls on source-of-truth data.

**Audit Trail:** View events and dossier exports logged.

**Integration with Other Modules:** Quick-add note → Knowledge Notebook; flag for follow-up → Workflow Centre.

### 8.2 Sections

| #  | Section            | Content                                                 |
|----|--------------------|---------------------------------------------------------|
| 1  | Customer Profile   | Demographics, KYC summary, contact, relationship since  |
| 2  | Loan Details       | Sanctioned/outstanding, ROI, tenure, repayment schedule |
| 3  | Documentation      | Document checklist with status flags                    |
| 4  | Risk Summary       | RHI, contributing rules, EWS triggers                   |
| 5  | Recovery           | Aging, Promise to Pay, follow-up history                |
| 6  | Legal              | Notices issued, case status, hearing schedule           |
| 7  | Inspection         | Last inspection, observations, ATR status               |
| 8  | Knowledge Notebook | Latest operational note + 'View all' link               |
| 9  | Timeline           | Chronological event feed                                |
| 10 | Attachments        | Read-only list of evidence files                        |

## 9. Knowledge Notebook

### 9.1. Knowledge Notebook 2.0

**Module Purpose:** Persistent operational memory layer for every account, surviving officer transitions.

**Navigation Path:** /branch/account/:accountId/notebook and /knowledge

**Accessible Roles:** Branch, Regional, Central (write requires officer role; read open to all permitted scopes)

**Business Objective:** Replace tribal knowledge and scattered spreadsheets with structured, searchable, versioned notes.

**Inputs:** 10 structured sections: Operational Notes, Inspection Notes, Recovery Notes, Customer Interaction, Legal Notes, Field Visit, Document Status, Restructure / OTS, Stress Indicators, General Remarks.

**Processing Logic:** Notes stored in a Zustand store with version history; each save appends an immutable revision with author, timestamp, and section tag.

**Validation Rules:** Section selection mandatory; minimum 10-character note body; attachments size capped.

**Outputs:** Section-wise note feed, Timeline view, Version History dialog, Officer Continuity handover pack.

**Reports Generated:** Notebook PDF export.

**Notifications:** Stress Indicator notes auto-raise an EWS alert.

**Dependencies:** Knowledge store, Account 360, EWS engine.

**Error Handling:** Optimistic save with rollback on failure; conflict resolution by latest-write-wins with full history retained.

**Security Controls:** Author identity stamped; older revisions immutable.

**Audit Trail:** Every save, edit, and view is logged.

**Integration with Other Modules:** Quick-Note dialog accessible from Account 360 and Workflow Centre; surfaces in Officer Continuity at /central/officers.

### 9.2 Officer Continuity

Generates a handover dossier listing all accounts owned by the outgoing officer, open follow-ups, last notes per account, and pending escalations. Available at /central/officers.

## 10. Rule-Based Risk Engine

### 10.1. Rule-Based Risk Engine

**Module Purpose:** Compute an explainable Risk Health Index (RHI) for every account, branch and region.

**Navigation Path:** /central/risk-intel

**Accessible Roles:** Central User, Admin

**Business Objective:** Make risk scoring transparent, configurable, and auditable.

**Inputs:** 13 rule inputs including DPD, EMI bounce count, security cover ratio, account turnover trend, CIBIL drift, restructure flag, sector stress, inspection overdue, legal exposure, CGTMSE coverage, document deficiency, EWS trigger count, geographic stress.

**Processing Logic:** Each rule produces a 0–100 sub-score; weighted sum normalised to 0–100. Bands: 0–40 High, 41–70 Medium, 71–100 Low risk.

**Validation Rules:** Weights must sum to 100; rule toggles auditable.

**Outputs:** RHI value, rule-contribution chart (Explainable Risk panel), risk band tag.

**Reports Generated:** Rule Configuration export, RHI distribution report.

**Notifications:** RHI band downgrade triggers EWS alert.

**Dependencies:** risk-intel-store, EWS engine.

**Error Handling:** Missing inputs fall back to neutral score with warning flag.

**Security Controls:** Rule editing limited to Admin role.

**Audit Trail:** Rule changes versioned with author and timestamp.

**Integration with Other Modules:** Consumed by Branch, Regional, Central dashboards and Account 360.

## 11. Early Warning System

### 11.1. Early Warning System

**Module Purpose:** Detect emerging risk and raise actionable alerts.

**Navigation Path:** Embedded in /branch, /central/risk-intel, and Account 360

**Accessible Roles:** All operational roles

**Business Objective:** Convert risk signals into work items with ownership and SLA.

**Inputs:** Rule outputs from the Risk Engine, manual stress-indicator notes, recovery missed-promises.

**Processing Logic:** Each trigger creates an alert with severity (High/Medium/Low), owner (branch officer by default), and SLA derived from configuration.

**Validation Rules:** Alerts deduplicated per account per rule within a 24h window.

**Outputs:** Alert list, severity badges, SLA countdown.

**Reports Generated:** Open EWS Alerts report.

**Notifications:** Dashboard badge + email automation simulator.

**Dependencies:** Risk Engine, Workflow Centre, Reminder Engine.

**Error Handling:** Resolved alerts archived, not deleted.

**Security Controls:** Resolution requires officer or above.

**Audit Trail:** Alert lifecycle (open → acknowledged → resolved) logged.

**Integration with Other Modules:** Resolution closes linked Workflow tasks.

## 12. Workflow & Task Management

### 12.1. Workflow Centre

**Module Purpose:** Unified queue and SLA management for all operational tasks.

**Navigation Path:** /workflow

**Accessible Roles:** All operational roles

**Business Objective:** Ensure every alert, follow-up, inspection action, and escalation has an owner, due date and audit trail.

**Inputs:** Tasks auto-generated from EWS, manual tasks from officers, escalations from RO/CO.

**Processing Logic:** 9 tab queues (My Queue, Branch Queue, Regional Queue, Escalations, Overdue, SLA Risk, Closed, Calendar, Manager Workbench). SLA computed from priority and due date.

**Validation Rules:** Task title, owner, due date mandatory; closure requires comment.

**Outputs:** Task lists, SLA dashboard, calendar view.

**Reports Generated:** Workflow Aging Report.

**Notifications:** Due-soon and overdue reminders via Reminder Engine.

**Dependencies:** workflow store, Reminder Engine, Email Automation simulator.

**Error Handling:** Optimistic UI with toast on save failure.

**Security Controls:** Reassignment restricted to managers.

**Audit Trail:** All state transitions logged.

**Integration with Other Modules:** Tasks link back to Account 360, Recovery, Inspection, Legal modules.



## 13. Reminder Engine

### 13.1. Reminder Engine

**Module Purpose:** Time-windowed reminders for tasks, follow-ups, inspections, legal hearings, and CGTMSE renewals.

**Navigation Path:** Embedded in Workflow Centre and Dashboard top bar

**Accessible Roles:** All

**Business Objective:** Reduce missed deadlines and SLA breaches.

**Inputs:** Task due date, priority, escalation matrix.

**Processing Logic:** Reminders triggered at T-7, T-3, T-1 and T+1 windows. Email Automation simulator renders preview of outbound emails.

**Validation Rules:** No reminders on closed tasks.

**Outputs:** Dashboard badge, in-app notification, email simulation card.

**Reports Generated:** Reminder Log report.

**Notifications:** In-app + simulated email.

**Dependencies:** Workflow store, notification panel.

**Error Handling:** Suppressed silently on duplicate windows.

**Security Controls:** User-scoped notifications.

**Audit Trail:** Reminder dispatch events logged.

**Integration with Other Modules:** Consumed by Workflow Centre, Branch Dashboard, Account 360.

## 14. Recovery Module

### 14.1. Recovery Module

**Module Purpose:** Track recovery actions, Promises to Pay (PTP) and follow-up history per overdue account.

**Navigation Path:** Account 360 → Recovery tab; aggregate at /branch (Recovery KPIs)

**Accessible Roles:** Branch User, Branch Manager, Regional User

**Business Objective:** Maintain a verifiable recovery trail and improve PTP conversion.

**Inputs:** PTP date, PTP amount, mode, contact channel, follow-up notes.

**Processing Logic:** PTP records aged into Honoured / Missed / Pending; missed PTPs auto-raise EWS alert.

**Validation Rules:** PTP date  $\geq$  today; amount  $> 0$ ; contact channel mandatory.

**Outputs:** PTP register, recovery history timeline, recovery vs target KPI.

**Reports Generated:** Recovery Performance Report (XLSX).

**Notifications:** Missed-PTP alert, due-PTP reminder.

**Dependencies:** EWS, Workflow, Reminder Engine.

**Error Handling:** Inline validation messages.

**Security Controls:** Branch scope enforced.

**Audit Trail:** Every PTP create/update logged.

**Integration with Other Modules:** Feeds Branch Dashboard KPIs and Regional ranking.

## 15. NPA Management

### 15.1. NPA Lifecycle

**Module Purpose:** Monitor SMA → NPA classification, provisioning indicators, and upgrade tracking.

**Navigation Path:** /central (NPA tab) and Account 360 → Risk Summary

**Accessible Roles:** Branch, Regional, Central

**Business Objective:** Provide consistent NPA visibility across all levels.

**Inputs:** Days past due, security cover, restructure status.

**Processing Logic:** Classifier maps DPD to SMA-0/1/2, Sub-Standard, Doubtful, Loss bands per RBI norms (mocked).

**Validation Rules:** DPD non-negative; classification immutable from source.

**Outputs:** NPA register, ageing analysis, upgrade pipeline.

**Reports Generated:** NPA Movement Report.

**Notifications:** Classification downgrade raises EWS.

**Dependencies:** Risk Engine, EWS.

**Error Handling:** Unknown DPD → 'Unclassified' with warning.

**Security Controls:** Read-only; no manual override.

**Audit Trail:** Classification changes logged.

**Integration with Other Modules:** Used by Risk Engine, Regional ranking, Central KPIs.

## 16. Legal Module

### 16.1. Legal Register

**Module Purpose:** Track legal notices, case filings, hearings, and status updates per account.

**Navigation Path:** Account 360 → Legal tab; aggregate at /central (Legal Register)

**Accessible Roles:** Branch, Central Legal

**Business Objective:** Centralise legal exposure visibility and hearing scheduling.

**Inputs:** Notice type, issue date, case number, court, next hearing, status, counsel.

**Processing Logic:** Hearings within 7 days surface in Workflow Calendar; status transitions tracked.

**Validation Rules:** Hearing date  $\geq$  today; status from controlled list.

**Outputs:** Case register, hearing calendar, status pie chart.

**Reports Generated:** Legal Exposure Report.

**Notifications:** Hearing reminders T-3, T-1.

**Dependencies:** Reminder Engine, Workflow Centre.

**Error Handling:** Past hearing dates flagged for reconciliation.

**Security Controls:** Confidential cases gated to legal role.

**Audit Trail:** All updates logged.

**Integration with Other Modules:** Feeds RHI rule weight and EWS.

## 17. CGTMSE Module

### 17.1. CGTMSE Management

**Module Purpose:** Track registration, coverage, claim status and renewal of CGTMSE-guaranteed accounts.

**Navigation Path:** /central (CGTMSE tab) and Account 360

**Accessible Roles:** Branch, Central

**Business Objective:** Prevent lapsed renewals and maximise claim recovery.

**Inputs:** CGTMSE reference, coverage amount, registration date, renewal date, claim status.

**Processing Logic:** Renewals within 60 days flagged; claim aging computed.

**Validation Rules:** Reference number unique; renewal date > registration date.

**Outputs:** CGTMSE register, renewal pipeline, claim aging.

**Reports Generated:** CGTMSE Coverage Report.

**Notifications:** Renewal reminders T-60/30/7.

**Dependencies:** Reminder Engine.

**Error Handling:** Duplicate reference rejected.

**Security Controls:** Branch scope enforced.

**Audit Trail:** Renewal and claim updates logged.

**Integration with Other Modules:** Feeds Branch KPI 'CGTMSE Coverage %'.

## 18. Inspection Module

### 18.1. Inspection & Audit

**Module Purpose:** Plan inspections, capture observations and track Action Taken Reports (ATR) to closure.

**Navigation Path:** /central (Inspection tab) and Account 360 → Inspection

**Accessible Roles:** Regional, Central, Inspectors

**Business Objective:** Drive inspection findings to verified closure within SLA.

**Inputs:** Inspection schedule, scope, observations, severity, ATR text, closure evidence.

**Processing Logic:** Each observation becomes a workflow task; closure requires ATR with reviewer approval.

**Validation Rules:** Severity mandatory; ATR required to close.

**Outputs:** Inspection register, observation list, ATR tracker.

**Reports Generated:** Inspection Closure Report.

**Notifications:** Overdue observations escalate per matrix.

**Dependencies:** Workflow Centre, Reminder Engine.

**Error Handling:** Closure without ATR blocked.

**Security Controls:** Reviewer role required for closure approval.

**Audit Trail:** Full observation lifecycle logged.

**Integration with Other Modules:** Feeds Central KPI 'Pending Inspections' and RHI.

## 19. Reports

All reports are read-only renderings of the underlying mock or uploaded data. Excel exports use SheetJS xlsx 0.20.3 (vulnerability-patched). Every export is watermarked with the SPBANKLENS 360 brand block defined in src/lib/export-brand.ts.

| Report               | Purpose                 | Inputs / Filters       | Output            | Export |
|----------------------|-------------------------|------------------------|-------------------|--------|
| Branch Daily Summary | Operational snapshot    | Branch (session), date | PDF               | PDF    |
| Regional Portfolio   | Branch ranking & risk   | Region, period         | On-screen + Excel | XLSX   |
| Executive MIS Pack   | Enterprise KPIs         | Zone, period           | Multi-sheet Excel | XLSX   |
| Recovery Performance | PTP and recovery        | Branch, period         | On-screen + Excel | XLSX   |
| NPA Movement         | Classification changes  | Period                 | On-screen + Excel | XLSX   |
| Legal Exposure       | Cases & hearings        | Branch / Region        | On-screen + Excel | XLSX   |
| CGTMSE Coverage      | Registration & renewals | Branch                 | On-screen + Excel | XLSX   |
| Inspection Closure   | Observations & ATRs     | Region, status         | On-screen + Excel | XLSX   |
| Open EWS Alerts      | Active alerts           | Severity, owner        | On-screen + Excel | XLSX   |
| Account Dossier      | Per-account 360         | Account ID             | PDF               | PDF    |

## 20. Administration

### 20.1. Administration

**Module Purpose:** Configure users, roles, risk thresholds, escalation SLAs and view audit logs.

**Navigation Path:** /admin

**Accessible Roles:** Security/Admin User

**Business Objective:** Centralise governance of the platform.

**Inputs:** User CRUD, role assignment, threshold values, SLA hours, retention policies.

**Processing Logic:** Changes versioned with author and timestamp; immediate effect on risk and workflow engines.

**Validation Rules:** Role mutually exclusive per user; thresholds within 0-100; SLA > 0 hours.

**Outputs:** User list, role matrix, configuration screens, audit log viewer.

**Reports Generated:** Configuration Change Log, User Activity Log.

**Notifications:** Configuration change broadcasts to active sessions.

**Dependencies:** Auth subsystem, Risk Engine, Workflow Centre.

**Error Handling:** Unauthorised attempts return 403.

**Security Controls:** Admin role required; double-confirm dialog on role escalation.

**Audit Trail:** Every admin action recorded with before/after values.

**Integration with Other Modules:** Affects every other module.



## 21. Search

### 21.1. Global Search

**Module Purpose:** Locate accounts, customers, branches, tasks and notes from a single top-bar input.

**Navigation Path:** App shell top bar

**Accessible Roles:** All

**Business Objective:** Reduce navigation time across modules.

**Inputs:** Free-text query (min 2 chars).

**Processing Logic:** Client-side index across mock entities with type filters (Account / Customer / Branch / Task / Note); ranked by recency and exact-match.

**Validation Rules:** Query length and scope respected.

**Outputs:** Grouped result list with deep links.

**Reports Generated:** n/a

**Notifications:** n/a

**Dependencies:** AppShell, all module stores.

**Error Handling:** Empty-state when no match.

**Security Controls:** Results filtered by user scope before render.

**Audit Trail:** Search events logged with hash of query.

**Integration with Other Modules:** Deep links into Account 360, Branch, Workflow.

## 22. Notifications

### 22.1. Notifications & Reminders

**Module Purpose:** Surface tasks, EWS alerts, reminders and escalations in a single panel.

**Navigation Path:** Top-bar bell icon (right) → Notification panel

**Accessible Roles:** All

**Business Objective:** Single inbox for everything that needs attention.

**Inputs:** Events from Workflow, EWS, Reminder Engine, Escalation Matrix.

**Processing Logic:** Notifications grouped by category (Task, Alert, Reminder, System); unread count badge; mark-as-read persists in session.

**Validation Rules:** Read state user-scoped.

**Outputs:** Notification list with action links.

**Reports Generated:** Notification Log (admin).

**Notifications:** Email Automation simulator card for email reminders.

**Dependencies:** workflow store, EWS engine.

**Error Handling:** Failed-to-load fallback message.

**Security Controls:** User-scoped; admin can view full log.

**Audit Trail:** Generated and read events logged.

**Integration with Other Modules:** Drives reminder badges across the app.

## 23. Security

### 23.1. Security Architecture

**Module Purpose:** Enforce authentication, authorisation, audit and read-only protection across the platform.

**Navigation Path:** Cross-cutting; settings at /admin.

**Accessible Roles:** All (enforcement), Admin (configuration).

**Business Objective:** Prevent privilege escalation, data tampering, and unauthorised disclosure.

**Inputs:** Session cookie, role claims, request context.

**Processing Logic:** Server-side role guard middleware on every protected route; httpOnly + Secure + SameSite session cookies; CSRF token on state-changing server functions; no client-trust for role.

**Validation Rules:** Role must match route requirement; session age within TTL.

**Outputs:** Allow / deny decisions; audit log entries.

**Reports Generated:** Audit Log report (Admin).

**Notifications:** Admin alerted on repeated failures.

**Dependencies:** Auth middleware, audit logger.

**Error Handling:** 401 / 403 with safe messages; no stack to client.

**Security Controls:** Read-only platform — no module writes to source systems. xlsx 0.20.3 (vuln-patched).

**Audit Trail:** All security-relevant events captured.

**Integration with Other Modules:** Applies to every other module.

## 24. Screen Specifications

This section catalogues the principal screens. For each screen the purpose, fields, buttons, validation, navigation and messages are listed. Screenshots are included in the companion SRS (v1.1) and Architecture decks.

### 24.1 Landing — Role Selection

#### Landing / Role Selection

**Purpose:** Allow the user to choose the portal appropriate to their role before login.

**Fields:** Role tiles (Branch, Regional, Central, Admin).

**Buttons:** Continue to Login.

**Validation:** A role must be selected before proceeding.

**Navigation:** → /login

**Error Messages:** 'Please select a role to continue.'

**Success Messages:** n/a

### 24.2 Login

#### Login

**Purpose:** Authenticate the user against server-issued session.

**Fields:** Username, Password (demo credentials panel visible).

**Buttons:** Sign In, Back to Role Selection.

**Validation:** Both fields required; server validates credentials and role.

**Navigation:** On success → role default dashboard.

**Error Messages:** 'Invalid credentials. Please try again.'

**Success Messages:** Silent redirect.

### 24.3 Branch Dashboard

#### Branch Dashboard

**Purpose:** Operational cockpit for branch staff.

**Fields:** KPI cards, date filter, segment filter, RHI gauge, charts.

**Buttons:** Export, Quick-Note, Open Workflow.

**Validation:** Date range ≤ 12 months.

**Navigation:** Drill-down to Account 360 / Workflow.

**Error Messages:** 'No data available for selected filters.'

**Success Messages:** 'Export generated successfully.'

### 24.4 Account 360

#### Account 360

**Purpose:** Consolidated account view across all source domains.

**Fields:** 10 tabbed sections; risk dial; EWS strip.

**Buttons:** Add Note, Flag for Follow-up, Download Dossier.

**Validation:** Account scope respected.

**Navigation:** Tab navigation; deep links from search.

**Error Messages:** 'Account not found or out of scope.'

**Success Messages:** 'Note saved.' / 'Task created.'

## 24.5 Knowledge Notebook

### Knowledge Notebook

**Purpose:** Structured note capture with version history.

**Fields:** Section selector, Note body, Attachments, Officer.

**Buttons:** Save, Save & Continue, View Versions.

**Validation:** Section +  $\geq 10$  char body required.

**Navigation:** Embedded in Account 360 and at /knowledge.

**Error Messages:** 'Please choose a section and enter at least 10 characters.'

**Success Messages:** 'Note saved (v{n}).'

## 24.6 Workflow Centre

### Workflow Centre

**Purpose:** Unified task and SLA management across queues.

**Fields:** Queue tabs, task table, due-date filter, owner filter.

**Buttons:** Create Task, Reassign, Close, Export.

**Validation:** Title, owner, due date mandatory.

**Navigation:** Deep links to source modules.

**Error Messages:** 'Cannot reassign — insufficient privilege.'

**Success Messages:** 'Task updated.'

## 24.7 Regional Dashboard

### Regional Dashboard

**Purpose:** Branch monitoring and escalation.

**Fields:** Heat map, ranking table, comparison panel.

**Buttons:** Compare, Drill to Branch, Export.

**Validation:** Region scope enforced.

**Navigation:** Drill-down to Branch Dashboard.

**Error Messages:** 'Branch data unavailable.'

**Success Messages:** 'Excel export ready.'

## 24.8 Central Executive Command Centre

### Executive Command Centre

**Purpose:** Enterprise governance view.

**Fields:** National KPIs, region heat map, sector treemap, ATR tracker.

**Buttons:** Drill, Export MIS Pack.

**Validation:** Period  $\leq 36$  months.

**Navigation:** Drill into Regional / Risk Intelligence.

**Error Messages:** 'Insufficient data for selected period.'

**Success Messages:** 'MIS Pack exported.'

## 24.9 Risk Intelligence Hub

### Risk Intelligence Hub

**Purpose:** Data upload, rule configuration, explainable scoring.

**Fields:** Data upload (XLSX/CSV), rule list, weight inputs, RHI preview.

**Buttons:** Upload, Apply Rules, Save Configuration.

**Validation:** File schema validated; weights sum 100.

**Navigation:** Located at /central/risk-intel.

**Error Messages:** 'Invalid file schema — see template.'

**Success Messages:** 'Configuration saved.'

## 24.10 Admin

### Administration

**Purpose:** User and configuration management.

**Fields:** User table, role assignment, threshold/SLA sliders, audit log viewer.

**Buttons:** Add User, Edit, Disable, Save Config.

**Validation:** Role + threshold + SLA constraints.

**Navigation:** /admin only.

**Error Messages:** 'Operation not permitted.'

**Success Messages:** 'Changes saved and broadcast.'

## 25. Functional Test Scenarios

| ID   | Scenario                 | Steps                                             | Expected                                                     |
|------|--------------------------|---------------------------------------------------|--------------------------------------------------------------|
| T-01 | Branch user login        | Select Branch role → enter credentials → submit   | Redirect to /branch with branch-scoped data                  |
| T-02 | Role guard               | Branch user navigates to /central via URL         | 403 / redirect to /branch                                    |
| T-03 | Account 360 access       | Open existing account ID in scope                 | All 10 sections render with mock data                        |
| T-04 | Notebook save            | Select section, type 15-char note, Save           | Note appears at top of feed; version v1 recorded             |
| T-05 | Notebook version history | Edit note, view version dialog                    | Previous revision retained with timestamps                   |
| T-06 | RHI explainability       | Open Account 360 → Risk Summary                   | Rule contributions chart sums to displayed RHI               |
| T-07 | EWS generation           | Configure rule to trigger; open Risk Intelligence | Alert appears with severity and SLA                          |
| T-08 | Workflow SLA             | Create task with due date T-1; wait               | Reminder badge appears; status flips to SLA Risk             |
| T-09 | Recovery missed PTP      | Mark PTP missed                                   | EWS alert raised; task created                               |
| T-10 | CGTMSE renewal reminder  | Set renewal date in 7 days                        | Reminder appears at T-7                                      |
| T-11 | Inspection closure block | Attempt close without ATR                         | Action blocked with validation error                         |
| T-12 | Regional drill-down      | Click branch on heat map                          | Branch Dashboard opens with that branch context              |
| T-13 | Central export           | Trigger MIS Pack export                           | Multi-sheet XLSX downloads with brand watermark              |
| T-14 | Admin role change        | Promote user to Manager                           | Audit log records change; user permissions update next login |
| T-15 | Logout                   | Click Logout                                      | Session cookie cleared; protected routes redirect to /login  |

## 26. Future Enhancements

The following capabilities are explicitly out of scope for v1.0 Prototype and are documented as a forward-looking roadmap:

- Live REST/GraphQL integration with CBS, LOS, recovery and inspection systems (replacing mock data).
- AI-assisted note summarisation and risk narrative generation.
- Predictive NPA model and behavioural scoring layered on top of the rule engine.
- Native mobile companion app for field officers (offline-first).
- SSO via the bank's enterprise identity provider (SAML/OIDC) and full MFA.
- Production-grade observability — metrics, traces, alerting, and SOC integration.
- Configurable report designer and scheduled report delivery via email.
- Multi-lingual UI.